

DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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PRELIMINARY EXAMINATION 2024 SECONDARY 4 NORMAL ACADEMIC

MATHEMATICS SYLLABUS A

4045/01

Paper 1

1 August 2024

2 hours

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Calculator Model

Answer **all** questions.

If working is needed for any question it must be shown with the answer.

Omission of essential working will result in loss of marks.

The total of the marks for this paper is 70.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π , use either your calculator value or 3.142, unless the question requires the answer in terms of π .

Total Marks

Mathematical Formulae*Compound interest*

$$\text{Total amount} = P \left(1 + \frac{r}{100} \right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3$$

$$\text{Area of a triangle } ABC = \frac{1}{2} ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Statistics

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f}$$

$$\text{Standard deviation} = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f} \right)^2}$$

Answer **all** the questions.

1 Write 0.0069084

(a) correct to 3 significant figures,

Answer [1]

(b) in standard form.

Answer [1]

2 Simplify $y^2 + 5xy + xy - 2y^2$.

Answer [2]

3 The number of visitors to a museum increased 16.4% between 2022 and 2023.
The number of visitors in 2023 was 97776.

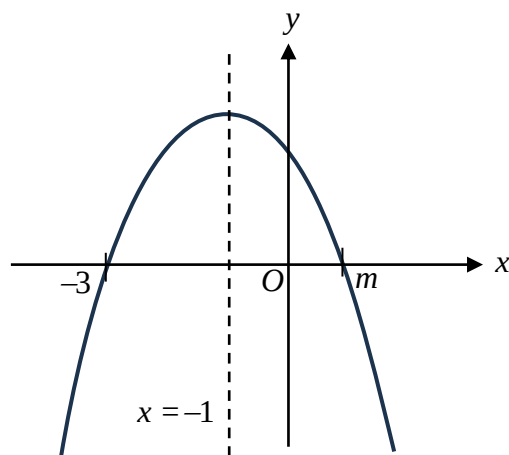
Calculate the number of visitors in 2022.

Answer [2]

4 Find obtuse angle A in radians such that $\sin A = 0.8$.

Answer $A = \dots\dots\dots$ radians [2]

- 5 The diagram shows the sketch of $y = k(x + 3)(x - m)$.
The line of symmetry of the graph is $x = -1$.



- (a) State the value of m .

Answer $m = \dots\dots\dots$ [1]

- (b) Write down a possible value of k .

Answer $k = \dots\dots\dots$ [1]

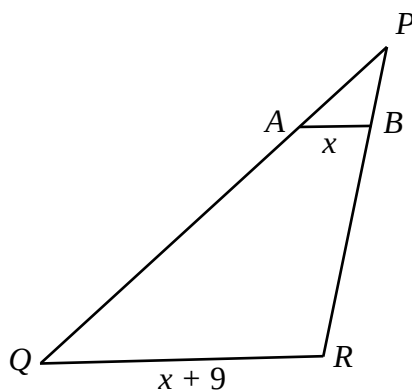
- 6 y is directly proportional to $\sqrt{x}$.

When $x = 9$, $y = 0.6$.

Find the value of y when $x = 100$.

Answer $y = \dots\dots\dots$ [2]

7



The triangles APB and QPR are similar.

$AB = x$ cm, $QR = (x + 9)$ cm and $\frac{PA}{AQ} = \frac{1}{3}$.

Find x .

Answer $x = \dots\dots\dots$ [3]

8 The stem-and-leaf diagram shows the number of goals scored by football teams last season.

0	3	6					
1	0	1	2	4	7	9	
2	3	7	9				
3	2	5	5	6	8		
4	0	4	5	9			

Key 2 | 3 means 23 goals

Find

(a) the number of teams that scored more than 35 goals,

Answer [1]

(b) the median.

Answer [1]

9 (a) Write $\frac{2}{x+3} + \frac{1}{x^2-9}$ as a single fraction in its simplest form.

Answer [2]

(b) Find the smallest prime number p satisfying $4p + 1 > 15$.

Answer $p =$ [2]

10 A world map is drawn to a scale of 1 : 25 000 000.

- (a) The distance between Bangkok and Tokyo is 18.4 cm.

Find the actual distance between Bangkok and Tokyo.
Give your answer in km.

Answer km [2]

- (b) Japan has a land area of 377973 km².

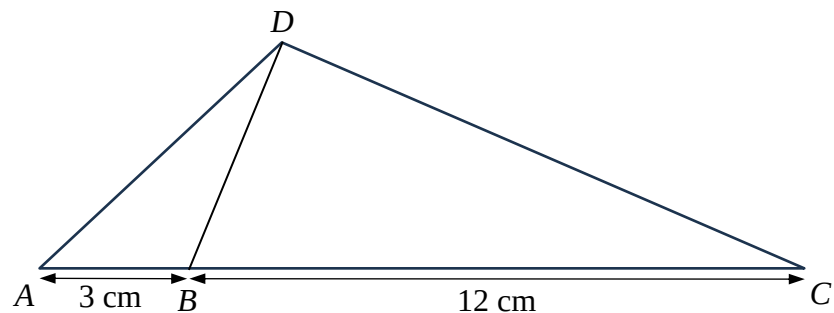
Calculate the area of Japan on the world map.
Give your answer in cm².

Answer cm² [2]

11 Solve $\frac{2x}{3} = \frac{4}{x-5}$.

Answer $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [3]

- 12 In the diagram, $AB = 3$ cm and $BC = 12$ cm.
 ABC is a straight line.



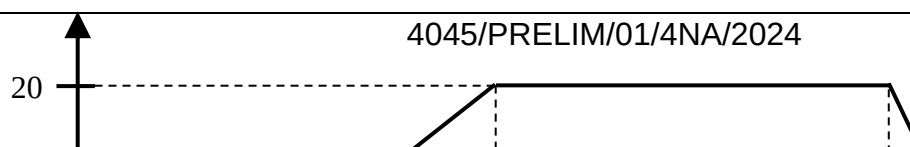
Sam says that the area of triangle BCD is 4 times the area of triangle ABD .

Is he correct? Explain your answer.

Answer $\dots\dots\dots$
 $\dots\dots\dots$
 $\dots\dots\dots$ [2]

- 13 The speed-time graph shows a journey.

Speed (m/s)



- (a) Calculate the acceleration.

Answer m/s^2 [2]

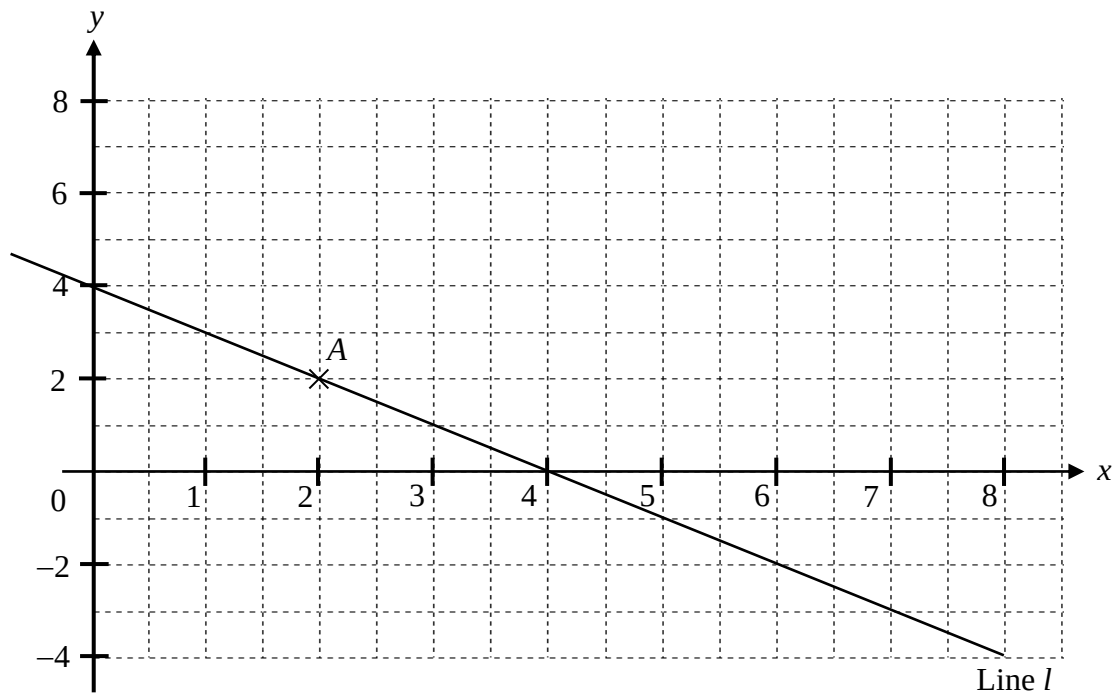
- (b) The area under the graph represents the total distance travelled.

Calculate the total distance travelled.

Answer m [2]

- 14** Part of the line l is drawn on the grid below.
A is the point (2, 2) and B is the point (n , 10).

A and B are points on the line l .



- (a) Find the value of n .

Answer $n = \dots\dots\dots$ [1]

- (b) Another line p passes through the points $(0, -2)$ and $(8, 6)$.

- (i) Draw the line p on the grid above. [1]

- (ii) State the coordinates of the point that lies on both line l and line p .

Answer $(\dots\dots\dots, \dots\dots\dots)$ [1]

15

$$x^2 + 10x + k = (x + a)^2$$

- (a) Find the value of a and the value of k .

Answer $a = \dots\dots\dots$

$k = \dots\dots\dots$ [2]

- (b) Hence solve $x^2 + 10x + k = 8$.
Give your answers correct to 2 decimal places.

Answer $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [2]

- 16 Point C lies on the line AB .
 $AC : AB = 7 : 11$.

- (a) Write down the ratio of $AC : BC$.

Answer $\dots\dots\dots : \dots\dots\dots$ [1]

- (b) AC is 6 cm longer than BC .

Calculate AB .

Answer $\dots\dots\dots$ cm [2]

- 17** The height, h , of an open cylindrical container is thrice its diameter.
The radius of the container is x cm.

(a) Write an expression for the height of the container, in terms of x .

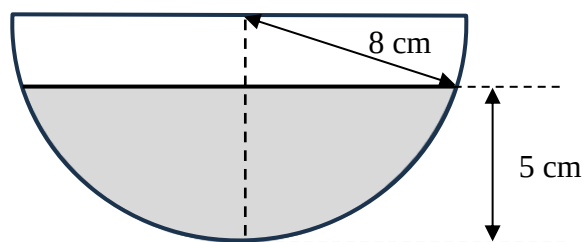
Answer $h = \dots\dots\dots$ [1]

(b) The total exterior surface area of the container is $208\pi \text{ cm}^2$.

Calculate x .

Answer $x = \dots\dots\dots$ [3]

18



The diagram shows the cross-section of a hemispherical bowl of radius 8 cm. Water is poured into the bowl to a depth of 5 cm.

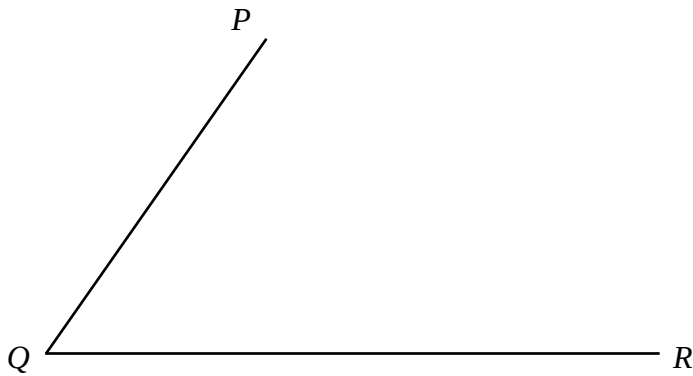
Find the area of the surface of the water.

Answer cm^2 [4]

19 The lines PQ and QR are shown in the space below.

(a) Construct parallelogram $PQRS$.

[1]



(b) On the parallelogram, construct

(i) the perpendicular bisector of QR ,

[1]

(ii) the bisector of angle PQR .

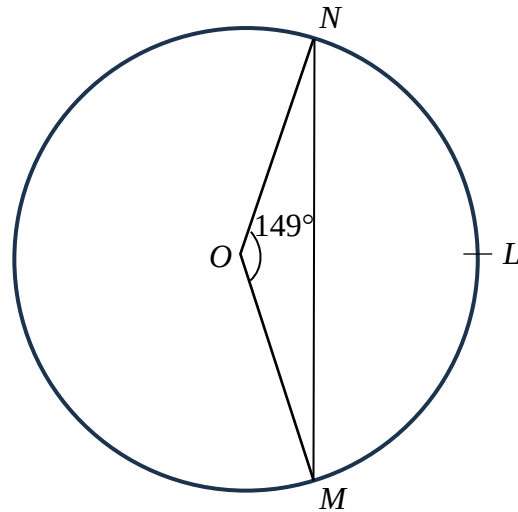
[1]

(c) A point K lies inside the parallelogram.

K is equidistant from Q and R , and equidistant from QR and PQ .

Mark and indicate clearly on the diagram, the position of K .

[1]



M and N are points on the circle centre O .

Angle $MON = 149^\circ$

The length of arc MLN is 18.2 cm.

Find the area of the major segment of the circle.

Answer cm^2 [5]

21 Solve these simultaneous equations.

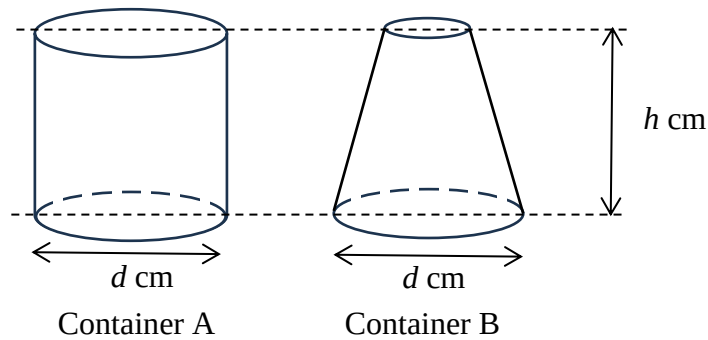
$$\begin{aligned}3x + 8y &= 11 \\4y &= 3 - x\end{aligned}$$

Show your working.

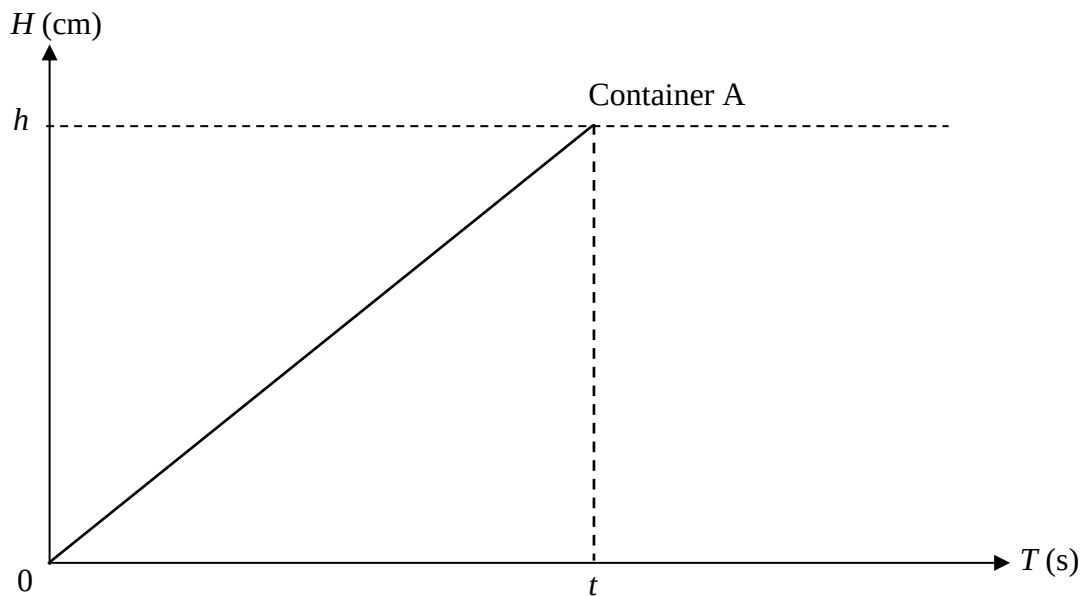
Answer $x = \dots\dots\dots$

$y = \dots\dots\dots$ [3]

- 22 The figure shows 2 empty containers with circular bases of diameter d cm. The containers are being filled to the brim with water at the same constant rate.



The diagram below shows the relationship between the height, H cm, of water and the time, T seconds, taken to fill the Container A.

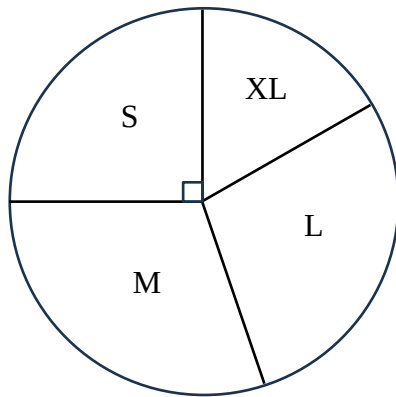


On the same diagram, sketch the relationship between the height, H cm, of water and the time, T seconds, taken to fill the Container B.

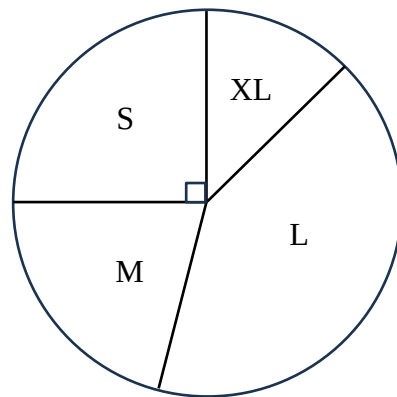
[2]

- 23 The sizes of Orientation T-shirts ordered in 2022 and 2023 are summarised on the accurate pie charts below.

Sizes of Orientation T-shirts ordered by year



Year 2022



Year 2023

- (a) Joe concludes that the number of S-sized Orientation T-shirts ordered in 2022 was the same as the number of S-sized Orientation T-shirts ordered in 2023.

State one feature of these pie charts that he has used to make this conclusion.
Explain why his conclusion may not be true.

Answer

.....

.....

..... [2]

- (b) The total number of Orientation T-shirts ordered in 2023 was 240.

Calculate the number of M-sized Orientation T-shirts ordered in 2023.

Answer T-shirts [2]

END OF PAPER